

JAMES PAOLO RILI

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OVERVIEW

Upcoming BCIT Computer Systems graduate and UBC Physics alumnus with 16 months of software engineering co-op experience and a strong foundation in physical data analysis. Proven ability in building modular software and obtaining insights from scientific data. Seeking full-time roles starting February 2027.

SKILLS

Programming Languages	Python, TypeScript, Bash, MATLAB, C, Rust
Data Analysis	NumPy, Matplotlib, Pandas, SciPy, scikit-learn, Keras, Jupyter
Tools & Methodologies	Git/GitHub, Linux, Agile, Atlassian Suite (Jira, Confluence)

WORK EXPERIENCE

Software Developer Co-op, Reporting Services Caracal Technologies <i>Vancouver, BC</i>	May 2025 - Dec 2025
- Engineered new features for an application processing high-volume (~10 million transactions/mo) transit data, delivering automated reports for clients using Python via AWS Lambda, EventBridge, and AuroraDB, as well as TypeScript and React.js. - Optimised 50+ SQL queries to fix performance issues, reducing data fetching times from over 3 minutes to under 10 seconds. - Streamlined workflow by configuring pipelines in BitBucket, writing Bash scripts, and documenting standard procedures in Confluence, reducing the risk of human error during deployment.	
RF Software Engineer Co-op, Hardware Validation NETGEAR, Inc. <i>Richmond, BC</i>	Sep 2022 - Apr 2023
- Developed and optimised Python-based software suite for WWAN hotspot device validation, ensuring that products are in compliance with the 3GPP standards while accelerating validation cycles by 50% through automation. - Built a Python library responsible for interfacing with a temperature chamber via RS-232, enabling automated temperature tests and minimising risk of human error during testing. - Added data visualisation with Matplotlib for live signal measurements, allowing testers to quickly analyse measurements and immediately pinpoint anomalies during validation.	
Physics Teaching Assistant University of British Columbia <i>Vancouver, BC</i>	Jan 2024 - Apr 2024
- Conducted weekly workshops for 50 undergraduate biology students to reinforce understanding of complex problems. - Delivered concise lectures about waves and hydrostatics to guide students' intuition of physical concepts in biology.	

PROJECTS

Machine Learning Interatomic Potentials UBC (https://github.com/jprili/gnn-mlip)	Apr 2024
- Engineered a neural-network model with Keras API and object-oriented design to accurately predict 3D crystalline atomic energies, providing a computationally efficient alternative to conventional, physics-based methods. - Leveraged Python's Matplotlib to provide visual intuition on the potential biases of the data and help provide starting points for optimising data cleanup. - Accelerated the search for the optimal model with feature engineering, cross-validation, and hyperparameter tuning.	
Analysis of Johnson-Nyquist Noise in Electrical Components UBC	Sep - Dec 2021
- Performed a series of experiments in an electrical laboratory investigating the effect of cosmic background noise on resistors. - Analysed data using NumPy and Pandas, and configured calculations for theoretical models with SciPy. - Composed a research paper detailing the process and presented the results to an audience of researchers and physicists.	
Iron-OBD2 BCIT + Iron IoT (https://github.com/ironiot/iron-obd2/)	Apr - May 2026
- Led a 5-person agile team in building a MVP of cloud-based OBD-II dashboard system with server-based data persistence. - Compiled documentation outlining system architecture, setup, and operating procedures in order to ensure a seamless handoff to the client and enable future development teams to easily maintain, scale, and troubleshoot the system. - Wrote unit-testing with Python and automation with Bash scripts, accelerating deployment while maintaining bug-free software.	

EDUCATION

Computer Systems Technology Diploma, Programming Paradigms British Columbia Institute of Technology	Sep 2024 - Dec 2026
Bachelor of Science in Physics University of British Columbia	Sep 2019 - Apr 2024